

# Global EIC Controls Software, Networking, & Computing PDR Presentation Planner

4/24/2026

Review Dates: June 15 - 17

## Charge Questions

1. Are the system requirements sufficiently defined, understood, and documented for this phase of the design?
2. Do the designs meet the requirements?
3. Are the interfaces sufficiently defined, understood and documented for this phase of the design?
4. Are the design analyses, simulations, drawings, specifications, and work plans sufficient for this phase of the design?
5. Have technical risks been identified and are mitigation plans adequate for this phase of the design?
6. Are the plans to address ES&H and Quality sufficient for this phase of the design?
7. Is the overall design maturity sufficient to proceed to the final design phase?

## Background

- Consider how we should speak to ESH and QA/QC topics in each talk... there will not be a dedicated Controls or ESH/QA talk on each
- Consider how Project risks and technical challenges are relevant to each talk
  - We have existing risks that can be highlighted
  - We should quickly consider adding risks to the Registry if they're warranted
- Emphasize that interfaces and related requirements are understood
  - Formally captured and tracked
  - Functional requirements should be mature for Preliminary Design stage
  - Performance requirements should be mostly mature for Preliminary Design
- General themes are...
  - Using community standard, open-source tools (ex. EPICS) where possible, rather than using custom approaches
  - EIC designs may change slightly, but the general plans (ex. Lattice, support systems) are well established
  - Engineering Verification Unit (EVU) demonstrations are either completed or are in progress where we think they are necessary prior to construction
    - Don't use the term "prototype"

- We design a modular and scalable system to adapt to changing needs over the length of the Project
  - Implementation tools and strategies will change, but our underlying strategies mitigate the Project Risks and related impacts and address our technical challenges

## Presentations

| Speaker | Topic                                     | Duration   | Notes                                                                                                                                                                                                                                                                                                |
|---------|-------------------------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Jim     | Welcome                                   | 5 mins     | <ul style="list-style-type: none"> <li>• Safety Briefing</li> <li>• Welcome &amp; Committee Intro</li> <li>• Etiquette</li> </ul>                                                                                                                                                                    |
| Daniel  | EIC Physics Overview                      | 15+5 mins  | <ul style="list-style-type: none"> <li>• EIC Physics background</li> <li>• Highlight requirements related to SW, Net, or Comp systems</li> <li>• Optimization needs</li> <li>• Reference similarities to RHIC operations for HSR</li> </ul>                                                          |
| ZY      | System Engineering Approach               | 15+5 mins  | <ul style="list-style-type: none"> <li>• General Systems Engineering processes (rather than an in-depth sampling of details)</li> </ul>                                                                                                                                                              |
| Jim     | Controls System Overview                  | 20+5 mins  | <ul style="list-style-type: none"> <li>• Team overview</li> <li>• In-scope Controls vs out-of-scope</li> <li>• Requirements sample (system?)</li> <li>• ESH/QC process background</li> <li>• General interfaces</li> <li>• Common Hardware Platform basics and connection to this PDR</li> </ul>     |
| Kunal   | Detailed Controls Architecture            | 25+10 mins | <ul style="list-style-type: none"> <li>• Detailed technical overview of the architecture (setup slide for mapping to other talks)</li> <li>• EPICS/ADO controls co-existence (Interfaces)</li> <li>• Requirements</li> <li>• Scalability</li> <li>• Integration (connect to Latif's talk)</li> </ul> |
| Kyle    | Facility Networking Architecture          | 15+5 mins  | <ul style="list-style-type: none"> <li>• Facility architecture</li> <li>• fiber plant</li> <li>• physical layout</li> <li>• Cybersecurity</li> </ul>                                                                                                                                                 |
| Kyle    | Controls Networking & Subnet Architecture | 15+5 mins  | <ul style="list-style-type: none"> <li>• Detailed architecture</li> <li>• Directory Service</li> <li>• Databases</li> <li>• ADO Bridge support (we can support the bridge tools)</li> </ul>                                                                                                          |

|        |                                         |           |                                                                                                                                                                                                                                                                                                                  |
|--------|-----------------------------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        |                                         |           | <ul style="list-style-type: none"> <li>• Gateway Services</li> <li>• Communication Interface</li> </ul>                                                                                                                                                                                                          |
| Zaiwen | Computing & Infrastructure Platform     | 15+5 mins | <ul style="list-style-type: none"> <li>• Architecture</li> <li>• Virtualization features</li> <li>• Storage</li> <li>• Deployments</li> <li>• OS</li> </ul>                                                                                                                                                      |
| Latif  | Controls DevOps & Infrastructure        | 15+5 mins | <ul style="list-style-type: none"> <li>• Dev Environment</li> <li>• CI/Release/Traceability</li> <li>• Project Support <ul style="list-style-type: none"> <li>○ eTraveler</li> <li>○ Component Data Base</li> </ul> </li> </ul>                                                                                  |
| Latif  | IOC Development & Hardware Integration  | 15+5 mins | <ul style="list-style-type: none"> <li>• EPICS Base plans</li> <li>• IOC Deployment infrastructure</li> <li>• Configuration management</li> <li>• Interface examples?</li> <li>• Standard modules to deploy</li> </ul>                                                                                           |
| Seth   | Middle Layer Services Architecture      | 15+5 mins | <ul style="list-style-type: none"> <li>• Sequencing</li> <li>• Alarms</li> <li>• Electronic Logbook</li> <li>• Save-Restore</li> <li>• Putlog, Autosave</li> <li>• Controls Restrictions</li> <li>• Workflows</li> <li>• Archiver (scalability)</li> <li>• Channel Finder / Name Service(scalability)</li> </ul> |
| Kunal  | Application Layer & Operator Interfaces | 15+5 mins | <ul style="list-style-type: none"> <li>• Operator Interface</li> <li>• Custom Applications</li> <li>• Client and utility libraries</li> </ul>                                                                                                                                                                    |
| Alexei | Commissioning Tools                     | 15+5 mins | <ul style="list-style-type: none"> <li>• Interface description between Commissioning and Controls</li> <li>• Requirements sampling</li> <li>• virtual accelerator plans</li> <li>• Consider application list?</li> </ul>                                                                                         |
| Jim    | Path Forward                            | 15+5 mins | <ul style="list-style-type: none"> <li>• Major tasks to reach FDR <ul style="list-style-type: none"> <li>○ ICD development</li> <li>○ demo/eval activities?</li> </ul> </li> </ul>                                                                                                                               |

## Documents for PDR

- IRD

- HLAs
  - Networking & Computing
- PRD
  - HLAs – ver0 approved
  - Networking & Computing
- General Functional Requirements
- Design Review Plan
  - One for HLAs + Networking & Computing
- QC Plan (draft only)
  - One shared QC Plan in progress
- RoDs
  - EPICS 7
  - Phoebus
  - GitHub
  - PVXS
  - Networking HW standard?
  - Virtual Computing standard?
  - ADO/EPICS Bridging Options?
- SOWs
  - Networking?
  - Computing?
- Specs
  - Networking?
  - Computing?
  - Commissioning tools?
- ICDs (draft only)
  - Consider linking to draft versions of existing building ICDs or similar documents
- Analyses, Simulations, Work Plans (consider)
  - Doc for ADO/EPICS bridging tool performance analysis?
    - Andrei & Latif?
  - Alarm bridging evaluation?
    - Seth?
  - Release Process document
    - Seth, Latif, & Kyle?